Wall time per step: Parse 1M PBMC, 500k (346,259 cells)

Seconds per step. Colour is scaled per step and size: fastest framework 0, slowest 1, on square−root seconds.

	bixverse mtx	bixverse h5ad	bixverse GPU*	bixverse MLX*	Scanpy mtx	Scanpy h5ad	Scanpy dask	scrapper mtx	scrapper h5ad	scrapper HDF5	Seurat std mtx	Seurat std h5ad	Seurat BPCells
load	46	25	21	21	OOM	12	23	OOM	19	372	OOM	OOM	71
qc	1.8	1.8	1.2	1.2	–	3.8	4.0	–	2.1	5.8	–	–	4.6
normalise	n/a	n/a	n/a	n/a	–	3.5	0.0	–	0.4	0.0	–	–	2.3
hvg	1.1	1.1	0.8	0.8	–	3.2	5.0	–	0.9	13	–	–	14
pca	2.2	2.0	1.1	0.8	–	5.7	9.8	–	14	50	–	–	352
harmony	2.1	2.3	1.3	0.8	–	2.4	2.3	–	122	125	–	–	89
knn_snn	12	12	5.8	5.4	–	31	27	–	27	28	–	–	31
cluster	4.0	4.1	3.3	3.3	–	14	13	–	85	92	–	–	104
cluster_fast	12	12	1.9	1.6	n/a	n/a	n/a	–	118	118	n/a	n/a	n/a
umap	22	22	2.6	2.5	–	155	153	–	126	126	–	–	198
dge	2.8	2.9	2.4	2.3	–	602	–	–	38	1,424	–	–	555
tsne_default	151	154	136	135	–	796	806	n/a	n/a	n/a	–	–	588
tsne_fft	39	38	9.4	14	–	109	109	n/a	n/a	n/a	–	–	106
tsne_quick	24	25	15	14	n/a	n/a	n/a	–	99	101	n/a	n/a	n/a
total (no tSNE)	94	73	39	38	failed	832	237	failed	434	2,237	failed	failed	1,421

Black: n/a (step not in that framework), OOM or TIMEOUT where the run died, − not reached.
Each step is scaled over the runs that finished it: at 1M the bixverse CPU tSNE is red because the others timed out.
Scanpy dask has no DGE step. bixverse GPU and MLX run natively without a memory cap.
bixverse CPU tSNE builds its kNN with IVF from 500k (kmknn at 100k). (qd) steps are not in the total.